

OPEN SOURCE And EMERGING PROTOCOLS Lead The Way For THE IoT



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Ever since the Internet of Things (IoT) came into the picture, it has caused one disruption after another. After all, the prospect of connecting anything non-computer to the Internet sounds exciting. But who knew that even after connecting your security system to the Internet to send out an alert to the authorities, you could still get robbed. Nevertheless, there is a lot of exciting stuff going on in the IoT spectrum.

T. Anand, principal architect, Knewron, says, "Newer communication protocols are helping in making wider-networked devices work together, without the need of heavy communication infrastructure." "These are more secure, consume less power and give a simplified interface to the Cloud and mobile phones," adds Akshay Aggarwal, co-founder, Mektrasy. Gartner, in its report about working areas in the coming two years, has highlighted low-power networks and security as important areas.

Focus areas in the IoT

Connectivity has seen a lot of focus over the past couple of years. Selecting a wire-

less network for an IoT device involves balancing multiple requirements including range, battery life, bandwidth, density, endpoint cost and operational cost. Multiple connectivity solutions like Long-Range Wide Area Network (LoRaWAN), Narrowband, Thread, Weightless or white space are some of the contenders. The long-term goal of an IoT network is to deliver data rates up to tens of kilobits per second (kbps) while consuming low power.

Low-power networks are on the rise.

"Many new transmission protocols are addressing range issues by keeping data rates lower," says Anand. LoRaWAN has been popular over the past couple of years in low-power networks. As per a test run in Bern, Switzerland, a single access point could connect to a device 49km away. With that much capacity, you could connect an entire country with a handful of such access points. This network, however, is more suited towards machine-to-machine (M2M) communication, and manufacturing plants seem to have realised the importance.

"We are seeing automobile manufacturers collaborating for LoRa usage," says Rishabh Chauhan, global community manager, The Things Network. He explains, "The trick lies in controlling the duty cycle and allowing maximum number of devices to use the network."

Phani Varanasi, chief executive officer, Hanbit Automation Technologies Pvt Ltd, looks at innovative uses of LoRa. "We are looking at using a form of global positioning system (GPS) that uses LoRa instead of General-Packet Radio Service (GPRS)." The user device pings off multiple access points and allows information on the location of the beacon at any point of time. "This is helpful in place like a large shipyard where it gets difficult to track individual vehicles over a couple of

Fig. 1: The number of connected devices will exceed 50 billion by 2020

